GTU ROBOTICS CLUB

Docker setup:

1.Start the basic knowledge of docker
2.Setup the docker in your system :
https://youtu.be/cqbh-RneBlk?si=WGxigQtl1LdN9a9W

3.Set the Nvidia - smi in your host system for gpu access
See the docs of Tensorrt_cuda_deployment_pdf in GITHUB

4.Set the nvidia-container toolkit
https://docs.ultralytics.com/guides/docker-quickstart/#prerequisites
Go through from this website
Verify NVIDIA-SMI Command
Start the setup of NVIDIA container toolkit from this website

5. Find the best docker page
LINK: https://hub.docker.com/r/ultralytics/ultralytics/tags
Example : Pull this image
Cmd: docker pull ultralytics/ultralytics:latest-python-export

6. Check through command
Cmd: **sudo docker images**(for images)
Cmd: **sudo docker ps -a**(for container)

7.enter the docker images with gpu , usb, photo, video,

Start with photo
Cmd : echo $DISPLAY
Cmd: # On host
xhost +local:root (for photo, gui access everytime run with or enter the container after above two cmd run)

8. Enter the setup of NVIDIA-SMI host system as well as then into the docker
A.cuda
B.Nvidia-smi
C.cudnn
D.Tensorrt

File from host to into the docker:
Cmd: sudo docker cp /home/purushottam/Downloads/nv-tensorrt-local-repo-ubuntu2204-10.13.2-cuda-13.0_1.0-1_amd64.deb ultralytics_container:/ultralytics/

9. Start this basic configuration
apt-get update
apt-get install -y usbutils
Check cmd: lsusb

Start the downloading and installing library
opencv -python, pyrealsense and other necessary libraries

<u>Install the nano for edit and make a new file</u>
apt-get update
apt-get install -y nano

Basic nano shortcuts

- **Ctrl + O** → Save the file (writes out)
- **Enter** → Confirm filename when saving
- **Ctrl + X** → Exit nano
- **Ctrl + K** → Cut a line
- **Ctrl + U** → Paste a line
- **Ctrl + W** → Search text

Go through this website for docker as well as host system for Realsense SDK

Link :

https://github.com/IntelRealSense/librealsense/blob/master/doc/distribution_linux.md

```
root@0694fdbd00f0:/ultralytics# lsusb
Bus 002 Device 011: ID 8086:0b5c Intel(R) RealSense(TM) Depth Camera 455  Intel(R) RealSense(TM) Dep
Bus 002 Device 001: ID 1d6b:0003 Linux 6.8.0-83-generic xhci-hcd xHCI Host Controller
Bus 001 Device 003: ID 048d:c966 ITE Tech. Inc. ITE Device(8176)
Bus 001 Device 002: ID 5986:212b SunplusIT Inc Integrated Camera
Bus 001 Device 004: ID 8087:0026
Bus 001 Device 001: ID 1d6b:0002 Linux 6.8.0-83-generic xhci-hcd xHCI Host Controller
root@0694fdbd00f0:/ultralytics# ls /dev/video*
/dev/video0  /dev/video1  /dev/video2  /dev/video3  /dev/video4  /dev/video5  /dev/video6  /dev/vid
root@0694fdbd00f0:/ultralytics# ls
2025-09-23-161706.webm  README.zh-CN.md                                                  docker
CITATION.cff            ai.jpeg                                                          docs
CONTRIBUTING.md         best.pt                                                          examples
LICENSE                 cuda-repo-ubuntu2204-13-0-local_13.0.1-580.82.07-1_amd64.deb     mkdocs.yml
README.md               cudnn-local-repo-ubuntu2204-9.13.0_1.0-1_amd64.deb               nv-tensorrt-l
root@0694fdbd00f0:/ultralytics# python3 realsense_test.py
root@0694fdbd00f0:/ultralytics# nano trained_model.py
root@0694fdbd00f0:/ultralytics# python3 trained_model.py

0: 480x800 (no detections), 49.7ms
Speed: 3.7ms preprocess, 49.7ms inference, 16.0ms postprocess per image at shape (1, 3, 480, 800)
```

Check above all command working in inside docker container and run your first code with ultimate GPU access.(plug intel realsense as well)